Passenger Preference Analysis in the Domestic Airline Industry - Business Insights and Relevance

Business Insights

1. Occupational Trends

The customer base of the airline under consideration comprises mainly Working Professionals and Students. Together, they make the largest segment of the customer base. About 700 customers are Working Professionals, and around 670 customers are students. This shows a division of the market between "Time-Critical" customers and "Budget-Critical" customers.

2. Age & Price Sensitivity

The age group between 25 and 34 years comprises the largest customer segment of the airline under consideration. This shows the peak age of professional mobility. The age groups 19-24 years and 35-44 years follow the largest customer segment. Price sensitivity is high among all the customers. More than 60% of all the age groups are "Very Sensitive" to the price of the tickets.

The age group 60+ years shows the highest price sensitivity among all the customers. About 67.4% of the customers falling under this age group are "Very Sensitive" to the price of the tickets. The age group 45-60 years shows the highest number of customers "Not Sensitive" to the price of the tickets. About 29.0% of the customers falling under this age group are "Not Sensitive".

3. Customer Segmentation (KMeans Clustering)

The clustering analysis shows the existence of four distinct types of travelers based on their behavioural and demographic characteristics. These are:

- Cluster 0 (Red) - Budget Multi-Purpose Travelers: This is the largest cluster of travelers. The customers have varied travel purposes. However, they are moderately price-sensitive.
- Cluster 1 (Purple) - Frequent Business Travelers: This cluster of customers shows distinct characteristics. The customers are highly frequency-oriented. They have specific schedule requirements.
- Cluster 2 (Brown) - Service Travelers: This cluster comprises customers who are highly focused on the quality of the services provided on the flight. They are highly satisfied with the quality of the flight services.
- Cluster 3 (Grey) - Price-Sensitive Leisure Travelers: This is the most price-sensitive segment. They are mainly focused on the lowest airfare.

4. Predictive Drivers of Airline Preference & Loyalty

The predictive model shows the fundamental factors driving the decision-making of the customers.

The Loyalty/Price Paradox

The predictive model shows that the Price Sensitivity factor dominates all other factors. This factor is the only driving force behind customer loyalty. This factor alone holds a significant weightage of more than 0.5. This shows the low brand loyalty of the customers. The brand loyalty of the customers is highly dependent on the price factor.

Predicting Airline Choice

Predicting Airline Loyalty: The initial decision to choose an airline is highly dependent on the Schedule Preference factor. This is followed by Travel Frequency. The customers are highly focused on the booking mode.

Carrier-Specific Influencers

- Akasa Air: This airline shows a high degree of influence from Ticket Price (25.6%).
- Vistara: This shows a high degree of influence from Safety Rating (16.8%).
- Air India & SpiceJet: These two airlines benefit the most from Loyalty Programs & Punctuality.

5. Customer Sentiment Trends

The sentiment analysis of over 23,000 customer reviews indicates that customer service is a dynamic area of focus. Out of all customer reviews, 54% are positive sentiment, whereas 45% are negative sentiment, indicating that service quality is a major determinant of brand preference.

6. Key Drivers of Customer Satisfaction

The text mining analysis on over 23,000 customer reviews indicates that satisfaction is built on a foundation of both operational efficiency and interpersonal warmth.

Interpersonal Excellence: Staff friendliness is the primary driver of positive sentiment, suggesting that human interaction is the most memorable aspect of the journey. When crew members go beyond basic service to show genuine care, it results in a disproportionately high positive impact on brand perception.

Physical Foundations: Seat comfort and cleanliness were identified as the "hygiene factors" of air travel. High-frequency positive mentions of cabin upkeep and anatomical support show that these are baseline requirements for a 5-star passenger experience.

Service Efficiency: Smooth and rapid processes, including check-in, baggage handling, and on-board dining service, act as immediate stress-reducers, correlating strongly with high satisfaction indices.

7. Major Causes of Customer Dissatisfaction

Operational instability and failures in professional conduct are the leading predictors of negative brand sentiment.

Operational Instability: Flight delays and cancellations are identified as the most damaging operational triggers. The predictive model shows that these issues not only cause immediate anger but also lead to long-term erosion of trust, outweighing other service factors.

Professional Conduct Failure: Staff rudeness is the singular most identified emotional trigger for customer dissatisfaction.

Coefficient Analysis: The specific terms "rude" and "unprofessional" emerged as the strongest coefficients in the dissatisfaction predictive model. This indicates a critical training gap, as passengers are significantly more likely to rate an experience poorly when they feel a lack of empathy or professional etiquette during stressful situations (like delays).

Business Relevance & Strategic Recommendations

1. Multi-Tiered Market Segmentation & Targeted Growth

The four behavioural clusters of Budget Multi-Purpose, Frequent Business, Service-Focused, and Price-Sensitive Leisure offer a marketing roadmap for high-impact strategies.

- Targeting the “Gender Growth” Gap :
  With males dominating 57% of passengers, there is a significant opportunity to increase female traveler participation. Airlines can introduce “Safety & Comfort” packages for leisure travel, targeting women who prefer flying in Premium Economy.

- Niche Capture for Business Owners :
  Since Business Owners have a 38% probability of choosing Business Class, airlines can introduce “Corporate Mobility” memberships targeting this niche group.

2. Dynamic Pricing & Revenue Optimization

The market is highly price-sensitive (60%+ of customers).

- Low Price Anchoring:
  Airlines must compete on price for the Price-Sensitive Leisure segment and Budget Seniors (60+). This can be done through Low Fare Alerts and Unbundled Economy options.

- Premium Upselling for the “Sweet Spot”:
  The 45–60 age group is the least price-sensitive segment (29% not price sensitive). Airlines can promote Premium Economy and Business Class services using value propositions like “Executive Comfort” and “Hassle-Free Travel.”

- Value-Based Scheduling:
  Since Schedule Preference is a key driver for airline selection, prime-time flights can be priced at a premium, especially for frequent business travelers.

3. Reimagining Loyalty in a Price-Dominant Market

The data clearly indicates a “Loyalty Paradox,” where price is the most important loyalty driver (>0.5 weight).

Therefore, traditional loyalty programs must evolve.

- Immediate Financial Gratification
  Introduce Instant Cashback or Companion Discounts instead of long-term point accumulation.
- Platform-Based Retention
  Introduce App-Only loyalty tiers to reduce dependence on third-party booking platforms.
- Encourage customers to book through the airline’s mobile application to receive exclusive loyalty benefits.

4. Operational Excellence & Brand Positioning

Text mining and predictive analysis indicate that operational issues strongly influence brand perception.

1. Zero-Tolerance for Operational “Rudeness”

Words such as “Rude” and “Unprofessional” strongly influence negative sentiment. Staff should be trained in emotional intelligence and customer interaction.

2. Competitive Differentiation

- Akasa Air: Focus on the Budget Multi-Purpose segment, as price dominance drives loyalty.
- Vistara: Leverage its strong safety rating (16.8%) to attract premium travelers.
- Air India & SpiceJet: Focus on punctuality and reliability, as these factors strongly influence customer loyalty.

5. Strategic Resource Allocation

- High ROI Service Areas
  Seat comfort, cleanliness, and staff friendliness are the top service areas influencing customer satisfaction. Investments should be directed toward these areas rather than brand promotion.
- Volatility Mitigation
  Since leisure travel is seasonal, airlines can stabilize demand by promoting student discounts and family group bookings, particularly targeting Education and Family segments (36%).